What is the best way to include R code in the Latex paragraph

I am a newbie in Latex. I have two simple questions regarding including R code in Latex.

Q1: For a block of R code, I use 'listings' package, as follows.

```
\documentclass[12pt]{report}
\usepackage{listings}
\begin{document}
\begin{lstlisting}[language=R]
> dpareto(4, 2, 3)
[1] 0.09375
> dpareto(4, -2, 3)
[1] NaN
> dpareto(seq(-1, 4, 1), 2, 3)
[1] 0.0000000 0.0000000 0.0000000 0.0000000 0.2962963 0.0937500
\end{lstlisting}
\end{document}
```

I feel it's not professional. What is the best way to present the block of R code?

Q2: For code pieces in the paragraph, for example I wish to include a R function, or variable, or package name in my Latex context, what is the best way to do that?

{r} {sourcecode}

asked Jan 30 '15 at 5:16
Aaron Zeng
198 1 3 9

1 Welcome to TeX.SX! Asking **one** question in a post is preferred ;-) – Christian Hupfer Jan 30 '15 at 5:24

Using `\lstinclude` you can include the code from external files, this could make sense in your case. Besides that I think that `knitR` or the older `Sweave` are *the* top tools to combine R and LaTeX. – Uwe Ziegenhagen Jan 30 '15 at 5:34

@ChristianHupfer, thanks for the comments. I feel my two questions are highly related. So I put them into one post. – Aaron Zeng Jan 30 '15 at 16:31

1 Answer

Better than list R code, you can make a `my_sweave_file.Rnw` file and compile it using `knitr` to execute the code, list it and show *real* results.


```
\documentclass[a4paper]{article}
\parindent0pt
\begin{document}
Knitr test in \LaTeX.

<<Test>>=

dpareto <- function(x, location, shape, log = FALSE) {
    if (!is.logical(log.arg <- log) || length(log) != 1)
        stop("bad input for argument 'log'")
    rm(log)

    L = max(length(x), length(location), length(shape))
    x = rep(x, length.out = L);
    location = rep(location, length.out = L);
    shape = rep(shape, length.out = L)

    logdensity = rep(log(0), length.out = L)
    xok = (x > location)
    logdensity[xok] = log(shape[xok]) +
            shape[xok] * log(location[xok]) -
            (shape[xok]+1) * log(x[xok])
    if (log.arg) logdensity else exp(logdensity)
}

dpareto(4, 2, 3)
dpareto(4, -2, 3)
dpareto(seq(-1, 4, 1), 2, 3)
@
Done.
\end{document}
```

To compile this .Rnw file you can do:

```
Rscript -e "library(knitr); knit('my_sweave_file.Rnw')"
pdflatex my_sweave_file.tex
```

But with editors like RStudio or LyX you can compile this files directly.

Note that the listed R code of you MWE is confusing and not reproducible, as you do not explain where `dpareto()` come from, so. If you use the library `PtProcess` instead of the listed function above, the result will be ratter different:

Knitr test in LaTeX.

```
library(PtProcess)
dpareto(4, 2, 3)

## [1] 0.28125

dpareto(4, -2, 3)

## Error in dpareto(4, -2, 3):  All values of lambda must be positive

dpareto(seq(-1, 4, 1), 2, 3)

## Error in dpareto(seq(-1, 4, 1), 2, 3):  Invalid data:  x < a
```

Done. The true first result is 0.28125, not 0.09375 as showed in the MWE.

```
\documentclass{article}
\parindent0pt
\begin{document}

Knitr test in \LaTeX.

{\footnotesize
<<Test, tidy=TRUE>>=
library(PtProcess)
dpareto(4, 2, 3)
dpareto(4, -2, 3)
dpareto(seq(-1, 4, 1), 2, 3)
@
}

Done. The true first result is \Sexpr{dpareto(4, 2, 3)}, not 0.09375
as showed in the MWE.

\end{document}
```

Or with the library `actuar` :

Knitr test in LaTeX.

```
library(actuar)

##
## Attaching package:  'actuar'
##
## The following object is masked from 'package:grDevices':
##
##     cm

dpareto(4, 2, 3)

## [1] 0.05247813

dpareto(4, -2, 3)

## Warning in dpareto(4, -2, 3):  NaNs produced

## [1] NaN

dpareto(seq(-1, 4, 1), 2, 3)

## Warning in dpareto(seq(-1, 4, 1), 2, 3):  NaNs produced

## [1] 0.00000000        NaN 0.28125000 0.14400000 0.08333333 0.05247813
```

Done. The true first result is 0.0524781, not 0.09375 as showed in the MWE.

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edited Jan 30 '15 at 7:47

answered Jan 30 '15 at 7:01

Fran
40.2k 6 92 144

1 Thanks a lot. That's really helpful. One follow-up question. So essentially I need first write .Rnw file, and

then compile it to .tex file, then to .pdf file. Is that correct? Additionally, if I wish to put R variable name or function name within a paragraph, what is the font/stype/format to do that? For example, I wish to say "here is the R function dpareto", then How I can format "dpareto" in the context to mark it as R code? – Aaron Zeng Jan 30 '15 at 16:37

That is, but with some editors is just push one bottom. Search also about `Sweave` (the still working ancestor of `knitr`). The last example show the two ways to include R code: between paragraphs with `<<>>=` and @ (that is like copy & paste chunks from the R console) and within paragraphs with `\Sexpr{}` that is formatted as plain LaTeX text. Read also about `xtable` and LaTeX ouput of `Hmisc` and others R packages here. – Fran Jan 30 '15 at 17:47